Energy Economics

LCC:-

i) It stands for Life Cycle Costing.

ii) It is defined as an important factor for comparing the alternatives and deciding on a particular process for completing a project.

iii) The different components taken into account for calculating LCC are:-

$$\boxed{LCC = \text{Capital Cost} + \text{Replacement Cost} + \text{Maintanance Cost} + \text{Energy Cost} - \text{Salvage}.}$$

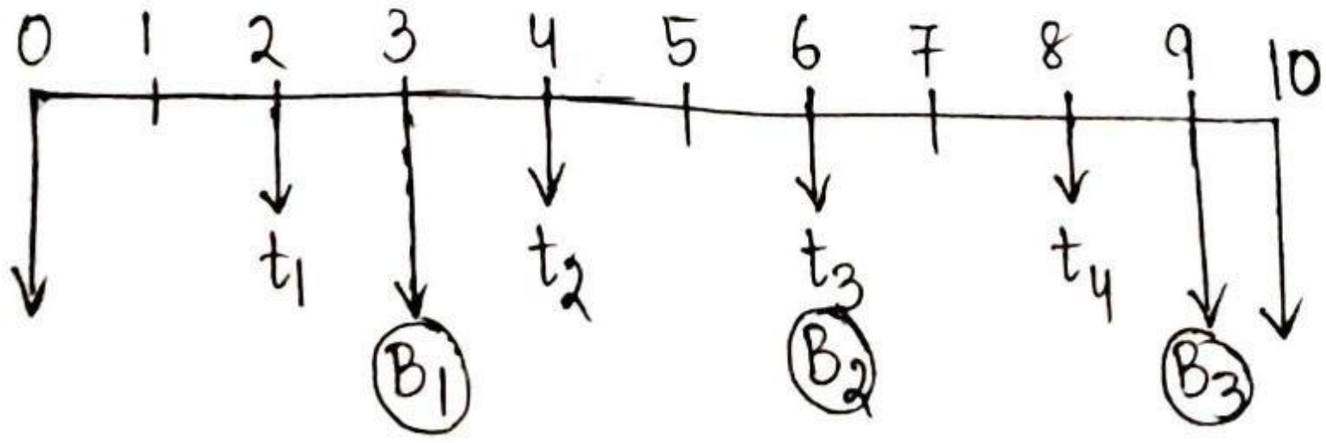
Steps to calculate LCC:-

i) Calculate the capital cost.

ii) Calculate the replacement cost

eg:- For a Bike, the lasting time for the components as:-

Tyres → 2 years, Brakes → 3 yrs, Lights → 1 yr



So,

$$\boxed{P_{\text{present worth}} = \frac{P_{\text{future worth}}}{(1+i)^n}}$$

where, i = Rate of interest

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iii) Calculate the maintanance cost.

$$\boxed{M_{\text{present worth}} = M_{\text{per yr}} \times \frac{1}{i}\left[1-\left(\frac{1}{1+i}\right)^n\right]}$$

At last,

$$\boxed{LCC = \text{Capital cost} + \text{Replacement cost} + \text{Maintanance cost} + \text{Miscellaneous cost}}$$

ALCC:-

i) It stands for Annuals Life Cycle Costing.

ii) It can be written as,

$$\boxed{ALCC = \frac{LCC}{\left(\frac{1+f}{i-f}\right)\left[1-\left(\frac{1+f}{i+1}\right)^n\right]}}$$

f = inflation rate.

↳ when, f = 0 :-

$$\boxed{ALCC = \frac{LCC}{\left(\frac{1}{i}\right)\left[1-\left(\frac{1}{1+i}\right)^n\right]}}$$

Q) A PV-array of 500 watts has been installed to pump water from a borewell of 2m deep using a submergable motor & pump system to an overhead tank. The length of pipe requiring to pump the water is 30 m. Following are the cost involved for the subsystems & their life spans :-

PV-array :- \$8/watt ⇒ Life span :- 15 yrs

[illegible] ⇒ Life span :- 7.5 yrs

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Pipe cost :- \$8/meter ⇒ life span :- 5 yrs
Cost of digging the bore well :- \$20/meter
Maintanance cost :- \$80/yr
Miscellaneous cost :- \$3.5/watt

If the interest rate is 10%, Calculate LCC for a period of 15 yrs & also calculate ALCC?

Ans:- Given,

$i = 10\%$, $n = 15$ yr

↳ i) Capital cost :-

Capital cost = cost of PV array + cost of Motor & Pump + Pipe cost + cost for digging the borewell

$\Rightarrow C_C = C_{PV} + C_M + C_P + C_B$

Here,

$C_{PV} = 500 \times 8\$ = (4000\$) = \$4000$

$C_M = 500 \times 2\$ = \$1000$

$C_P = 30 \times 8\$ = \$240$

$C_B = 2 \times 20\$ = \$40$

So,

$C_C = 4000 + 1000 + 240 + 40 \Rightarrow \boxed{C_C = \$5280}$

↳ Replacement Cost :-

Here,

$R_C = R_M + R_{pipe}$

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Now,

$$R_M = \frac{1000}{(1+0.1)^{7.5}} = \$\,489.27$$

$$R_{pipe} = \frac{240}{(1+0.1)^5} = \$\,149.02 \longrightarrow \text{for 1st replacement}$$

$$= \frac{240}{(1+0.1)^{10}} = \$\,92.53 \longrightarrow \text{for 2nd replacement}$$

So,

$$R_C = R_M + R_{pipe} = (489.27 + 149.02 + 92.53)$$

$$\Rightarrow \boxed{R_C = \$\,730.82}$$

↳ iii) Maintanance cost :-

Given, Maintanance cost/yr = M_C/yr = \$ 80/yr

We know,

$$M_{C\,present} = M_C/yr \times \frac{1}{i}\left[1 - \left(\frac{1}{1+i}\right)^n\right]$$

$$\Rightarrow M_{C\,present} = 80 \times \frac{1}{0.1}\left[1 - \left(\frac{1}{1.1}\right)^{15}\right]$$

$$\Rightarrow \boxed{M_C = \$\,608.48}$$

Now, we know that,

$$LCC = C_C + R_C + M_C + (\text{Miscellaneous cost})$$

$$\Rightarrow LCC = \$(5280 + 730.82 + 608.48) + [\$\,3.5/\text{watt}]$$

$$\Rightarrow LCC = \$(5280 + 730.82 + 608.48) + [\$\,3.5 \times 500]$$

$$\Rightarrow \boxed{LCC = \$\,8368.486}$$

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Now,

$$ALCC = \frac{LCC}{\left(\frac{1}{i}\right)\left[1-\left(\frac{1}{1+i}\right)^n\right]} = \frac{\$8368.486}{\left(\frac{1}{0.1}\right)\left[1-\left(\frac{1}{1.1}\right)^{15}\right]}$$

$$\Rightarrow \boxed{ALCC = \$1100.23} \text{ (Ans)}$$

Q) A community has 500 people. The source of water to the community is from the borewell and the supply of water from the borewells is by Hand-pumps. 6-Handpumps are installed to meet the water requirement of the community. Per capita water consumption of the community is 40 lit/day. Borewell depth is 20m, the cost of each handpump is 5000/-. The cost of digging borewell is at the rate of 250/- per meter. Lifespan of handpumps is 10 years, Annual maintanance cost per pump is 1250/-. If the rate of interest is 10%, what is the unit water cost for the life cycle period of 20 years?

Ans: Given, $i = 10\% = 0.1$, $n = 20$ yrs.

i) Capital Cost :-

C_c = Cost of Hand pump + Cost for digging borewells

$$\Rightarrow C_c = (6 \times 5000) + (250 \times 20 \times 6)$$

$$\Rightarrow \cancel{C_c = 350,000/-} \Rightarrow \boxed{C_c = 60,000/-}$$

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ii) Replacement Cost:-

$$R_C = R_{hand\ pump} = R_H$$

$$P_1 = \frac{30000}{(1+0.1)^{10}} = \boxed{11566.29/-} = R_C$$

iii) Maintanance cost:-

$$M_C = (1250 \times 6) = \boxed{7500/-}$$

we know,

$$M_{C_{present}} = M_C/yr \times \frac{1}{i}\left[1-\left(\frac{1}{1+i}\right)^n\right]$$

$$\Rightarrow M_{C_{present}} = 7500 \times \frac{1}{0.1}\left[1-\left(\frac{1}{1.1}\right)^{20}\right]$$

$$\Rightarrow \boxed{M_C = 63851.72/-}$$

We know that,

$$LCC = C_C + R_C + M_C$$

$$\Rightarrow \boxed{LCC = 135418.01/-}$$

Now,

$$ALCC = \frac{LCC}{\left(\frac{1}{i}\right)\left[1-\left(\frac{1}{1+i}\right)^n\right]} = \frac{135418.01/-}{\frac{1}{0.1}\left[1-\left(\frac{1}{1.1}\right)^{20}\right]}$$

$$\Rightarrow \boxed{ALCC = 15906.15/-}$$

So,

Annual water consumption in the community $= (40 \times 365 \times 50$

$= 730000$ lit/y

Now,

15906.15

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(94)

Q) The measure component of Biogas plant are listed as follows:- (i) Gasifier (ii) Piping (iii) Sand filter (iv) Diesel Engine (v) Electric generator

The cost of different components of biogas plant are specified in the following table:-

Item	Cost in Rs/- for 5 Kw plant	Life period
1) Biogas	127700	20 yr
2) Piping	8300	10 yr
3) Sand filter	4150	10 yr
4) 7HP diesel eng	37700	15 yr
5) 5KVA Generator	78150	15 yr
6) Accessories	20750	10 yr
7) Engine Room	16550	20 yr

If the rate of interest is 10% & the maintanance cost is 1% of (Total capital cost) per year. Calculate the LCC & ALCC

Ans: Given, $i = 10\% = 0.1$, $n = 20$ yr, $\cos\phi = 1$ (let)

i) Capital cost:-

$$C_C = C_B + C_P + C_{SF} + S_{DE} + S_G + S_A + S_{ER}$$

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$$\Rightarrow C_C = 127700 + 8300 + 4150 + \underbrace{\left[\frac{37700}{5000} \times (7 \times 746)\right]}_{39373.88} + 78150 +$$

$$20750 + 16550$$

$$\Rightarrow \boxed{C_C = 294973.88/-}$$

ii) <u>Replacement Cost:-</u>

$$R_C = R_P + R_{SF} + R_{DE} + R_G + R_A$$

$$R_P = \frac{8300}{(1+0.1)^{10}} = 3200/-$$

$$R_{SF} = \frac{4150}{(1+0.1)^{10}} = 1600/-$$

$$R_{DE} = \frac{39373.88}{(1+0.1)^{15}} = 9425.79/-$$

$$R_G = \frac{78150}{(1+0.1)^{15}} = 18708.48/-$$

$$R_A = \frac{20750}{(1+0.1)^{10}} = 8000/-$$

so, $\boxed{R_C = 40934.27/-}$

iii) <u>Maintanance cost:-</u>

$$M_C = 1\% \cdot (C_C) = 1\% \cdot (294973.88) = 2949.7388/-$$

$$M_{C\,present} = M_C/yr \times \frac{1}{i}\left[1 - \left(\frac{1}{1+i}\right)^n\right]$$

$$\Rightarrow M_{C\,present} = (2949.7388) \times \frac{1}{0.1}\left[1 - \left(\frac{1}{1.1}\right)^{20}\right]$$

$$\Rightarrow \boxed{M_{C\,present} = 25112.78/-}$$

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We know that,

$$LCC = C_C + R_C + M_C$$

$$\Rightarrow \boxed{LCC = 3,61,020.93/-}$$

Now,

$$ALCC = \frac{LCC}{\frac{1}{i}\left[1-\left(\frac{1}{1+i}\right)^n\right]} = \frac{361020.93}{\frac{1}{0.1}\left[1-\left(\frac{1}{1.1}\right)^{20}\right]}$$

$$\Rightarrow \boxed{ALCC = 42405.38297/-}$$

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